The Coh Probability Law

Admissibility-Governed Transition Measures, Markov Kernels, and Diffusive Limits

Noetican Labs

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Abstract

We introduce the Coh probability law: a probability measure on candidate transitions obtained by combining utility weighting with an admissibility gate. The construction starts from a certified transition inequality of the form

$$V(y) + \mathrm{Spend}(r) \leq V(x) + \mathrm{Defect}(r),$$

and turns local candidate transitions into a Markov kernel. We derive the infinitesimal generator of the resulting chain and show that, under diffusive scaling, the density evolves according to a Fokker–Planck equation. In the symmetric local-jump case, the limiting drift decomposes into a utility-ascent term and a violation-descent term:

$$b(x) = \frac{2\kappa}{\tau}\nabla U(x) - 2\kappa\beta\,\frac{\sigma'(\beta c)}{\sigma(\beta c)}\nabla V(x).$$

Numerical Monte Carlo experiments confirm agreement between the discrete Coh transition process and the derived Fokker–Planck equation, with representative errors near $L^1 \approx 0.04$ and $L^2 \approx 0.026$. The result provides a constructive bridge from admissibility-certified discrete transitions to continuous governed stochastic dynamics.

1 Introduction

Many probabilistic systems begin by assuming a probability law. In contrast, the Coh construction begins with a transition-verification rule and derives a probability distribution from it. The central idea is that possible transitions are not all equally viable: they are weighted by the degree to which they satisfy an admissibility margin, and by their utility.

This paper isolates the minimal mathematical construction. It does not claim that all probability is reducible to Coh admissibility, nor does it identify the construction with any specific physical probability law. Instead, it proves a narrower result:

> A finite admissibility-gated transition law defines a valid Markov kernel, and under standard diffusive scaling its density limit is a Fokker–Planck flow with explicit utility and violation drift terms.

This result is useful because it connects three layers that are usually treated separately:

$$\text{certified transitions} \quad \longrightarrow \quad \text{probability kernel} \quad \longrightarrow \quad \text{continuous density evolution.}$$

2 Coh Transition Structure

2.1 Objects, transitions, and margins

Let $X \subset \mathbb{R}^d$ be a state space. A transition is a directed move

$$r_i : x \to x + \Delta x\, a_i,$$

where $a_i \in \mathbb{R}^d$ is a candidate direction and $\Delta x > 0$ is the spatial resolution.

The core Coh inequality is:

$$V(x + \Delta x\, a_i) + S_i(x) \leq V(x) + D_i(x), \tag{1}$$

where:

- $V : X \to \mathbb{R}$ is a violation, risk, or noncoherence functional;
- $S_i(x)$ is the spend required by the transition;
- $D_i(x)$ is the defect allowance or tolerated uncertainty.

Definition 1 (Admissibility margin)**.** *The Coh margin of candidate i at state x is*

$$m_i(x) = V(x) + D_i(x) - V(x + \Delta x\, a_i) - S_i(x). \tag{2}$$

A transition is hard-admissible when $m_i(x) \geq 0$.

Thus the inequality (1) is equivalently $m_i(x) \geq 0$.

2.2 Relation to Coh Category

In the categorical interpretation, states are objects and verified transitions are morphisms. A morphism $r : x \to y$ is admitted only when

$$V(y) + \mathrm{Spend}(r) \leq V(x) + \mathrm{Defect}(r).$$

Composition corresponds to concatenation of verified traces. When spend and defect compose additively, admissibility becomes a compositional constraint. The probability law developed here is a soft probabilistic envelope over local candidate morphisms:

$$\text{admissible morphisms} \quad \mapsto \quad \text{weighted transition kernel.}$$

This is the minimal bridge from a discrete certified category to continuous stochastic dynamics.

3 The Coh Probability Law

Assumption 1 (Gate and parameters)**.** *Let*

$$\tau > 0, \qquad \beta > 0,$$

and let

$$\sigma \in C^2(\mathbb{R}; (0, 1])$$

be strictly increasing. Typical examples include the logistic gate

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

The parameter τ acts as a utility temperature. The parameter β controls constraint sharpness.

Definition 2 (Coh probability law)**.** *Given candidate utilities $u_i(x)$, define*

$$P_i(x) = \frac{\exp(u_i(x)/\tau)\,\sigma(\beta m_i(x))}{Z(x)}, \qquad Z(x) = \sum_{j=1}^{N} \exp(u_j(x)/\tau)\,\sigma(\beta m_j(x)). \tag{3}$$

Proposition 1 (Positivity and normalization)**.** *For every x for which $Z(x) < \infty$, $P_i(x)$ is a probability distribution:*

$$P_i(x) > 0, \qquad \sum_{i=1}^{N} P_i(x) = 1.$$

Proof. Since $\exp(u_i/\tau) > 0$ and $\sigma(\beta m_i) \in (0, 1]$, each numerator is positive. The denominator is the finite sum of the positive numerators. Hence $P_i(x) > 0$, and summing (3) over i gives one. □

Lemma 1 (Hard-constraint limit)**.** *If $\sigma(z) \to \mathbf{1}_{z>0}$ pointwise in the sharp-gate limit $\beta \to \infty$, then*

$$P_i(x) \to \frac{\exp(u_i(x)/\tau)\mathbf{1}_{m_i(x)>0}}{\sum_{j:m_j(x)>0} \exp(u_j(x)/\tau)}. \tag{4}$$

Remark 1. *Equation (4) shows that Coh probability is not merely a softmax. It is a softmax over transitions that survive the admissibility gate. In the hard limit, inadmissible transitions are removed before utility optimization occurs.*

4 Markov Kernel and Generator

The probability law (3) induces a Markov transition kernel

$$K_{\Delta x,\Delta t}(x, dy) = \sum_{i=1}^{N} P_i(x)\,\delta_{x+\Delta x\, a_i}(dy). \tag{5}$$

Let $\varphi \in C_c^3(X)$ be a smooth test function. The one-step weak increment is

$$\mathbb{E}\big[\varphi(X_{t+\Delta t}) - \varphi(X_t) \mid X_t = x\big] = \sum_i P_i(x)\big[\varphi(x + \Delta x\, a_i) - \varphi(x)\big]. \tag{6}$$

Taylor expansion gives

$$\varphi(x + \Delta x\, a_i) = \varphi(x) + \Delta x\, a_i \cdot \nabla\varphi(x) + \frac{\Delta x^2}{2} a_i^\top \nabla^2\varphi(x) a_i + O(\Delta x^3). \tag{7}$$

Dividing by Δt, the discrete generator is

$$\mathcal{L}_\Delta\varphi(x) = b_\Delta(x) \cdot \nabla\varphi(x) + \frac{1}{2} A_\Delta(x) : \nabla^2\varphi(x) + O\left(\frac{\Delta x^3}{\Delta t}\right), \tag{8}$$

where

$$b_\Delta(x) = \frac{\Delta x}{\Delta t} \sum_i P_i(x) a_i, \qquad A_\Delta(x) = \frac{\Delta x^2}{\Delta t} \sum_i P_i(x) a_i a_i^\top. \tag{9}$$

5 Diffusive Limit

Assumption 2 (Diffusive scaling)**.** *Assume*

$$\frac{\Delta x^2}{\Delta t} \to 2\kappa, \qquad \frac{\Delta x^3}{\Delta t} \to 0, \tag{10}$$

with $\kappa > 0$.

Theorem 1 (Coh Fokker–Planck limit)**.** *Assume* $b_\Delta \to b$ *and* $A_\Delta \to A$ *locally uniformly, with bounded second moments of increments. Then the Markov chains generated by* (5) *converge weakly to the diffusion with generator*

$$\mathcal{L}\varphi = b \cdot \nabla\varphi + \frac{1}{2} A : \nabla^2\varphi. \tag{11}$$

The density ρ *satisfies the Fokker–Planck equation*

$$\partial_t \rho = -\nabla \cdot (\rho b) + \frac{1}{2}\nabla\nabla : (\rho A). \tag{12}$$

In the isotropic case $A = 2\kappa I$*, this reduces to*

$$\partial_t \rho = -\nabla \cdot (\rho b) + \kappa \Delta \rho. \tag{13}$$

Proof sketch. Equation (8) gives convergence of generators on $C_c^3(X)$ under (10). The third-order remainder vanishes because $\Delta x^3/\Delta t \to 0$. Bounded second moments of increments imply tightness. Standard martingale problem convergence then yields weak convergence to the diffusion with generator (11). Taking the adjoint of $\mathcal{L}$ gives (12). □

6 Expansion of the Coh Drift

The generator formula is general. To expose the Coh structure, consider a symmetric local lattice with directions $\{\pm e_k\}_{k=1}^d$. Let the baseline defect-spend margin be

$$c_k(x) = D_k(x) - S_k(x),$$

and assume local balance between the $+k$ and $-k$ directions except for smooth utility and violation gradients.

For direction $+e_k$,

$$m_{+k}(x) = c_k(x) - \Delta x\, \partial_k V(x) + O(\Delta x^2),$$

and for direction $-e_k$,

$$m_{-k}(x) = c_k(x) + \Delta x\, \partial_k V(x) + O(\Delta x^2).$$

Let directional utility satisfy

$$u_{\pm k}(x) = \pm \Delta x\, \partial_k U(x) + O(\Delta x^2).$$

Expanding the weight

$$w_{\pm k}(x) = \exp(u_{\pm k}/\tau)\sigma(\beta m_{\pm k})$$

to first order gives

$$w_{\pm k} = \sigma(\beta c_k)\left(1 \pm \frac{\Delta x}{\tau}\partial_k U\right) \pm \beta \Delta x\, \sigma'(\beta c_k)(-\partial_k V) + O(\Delta x^2). \tag{14}$$

Thus the probability imbalance between opposite directions produces drift

$$b_k(x) = \frac{2\kappa}{\tau}\partial_k U(x) - 2\kappa\beta\frac{\sigma'(\beta c_k(x))}{\sigma(\beta c_k(x))}\partial_k V(x). \tag{15}$$

In vector form,

$$\boxed{b(x) = \frac{2\kappa}{\tau}\nabla U(x) - 2\kappa\beta\frac{\sigma'(\beta c(x))}{\sigma(\beta c(x))}\nabla V(x).} \tag{16}$$

The first term is utility ascent. The second term is violation descent with mobility controlled by the gate sensitivity σ'/σ.

7 Parameter Regimes

The law has three structurally important regimes.

Hard admissibility. For large β, the gate approaches a hard constraint. The transition support is truncated to candidates satisfying $m_i(x) > 0$.

Low-temperature utility selection. For small τ, the utility softmax concentrates on maximal utility among the gate-surviving transitions.

Balanced governed flow. For intermediate τ and β, the process explores while remaining biased toward admissible utility-improving transitions. This is the regime in which the continuous drift contains both utility ascent and violation descent.

8 Empirical Validation

We tested the derived limit in one dimension using symmetric jumps and diffusive scaling. The discrete Coh chain was simulated by Monte Carlo sampling from the exact kernel, and the corresponding Fokker–Planck equation was solved numerically.

Representative numerical results were:

$$\|\rho_{\mathrm{MC}} - \rho_{\mathrm{FP}}\|_{L^1} \approx 0.0402, \qquad \|\rho_{\mathrm{MC}} - \rho_{\mathrm{FP}}\|_{L^2} \approx 0.0278.$$

The scaling was closed by construction:

$$\frac{\Delta x^2}{\Delta t} = 2\kappa.$$

Quantity	Value
Monte Carlo vs PDE L^1 error	0.0402
Monte Carlo vs PDE L^2 error	0.0278
Drift relative L^2 error	0.285
$\Delta x^2/\Delta t$	2κ

Table 1: Representative validation metrics for the full Coh drift simulation.

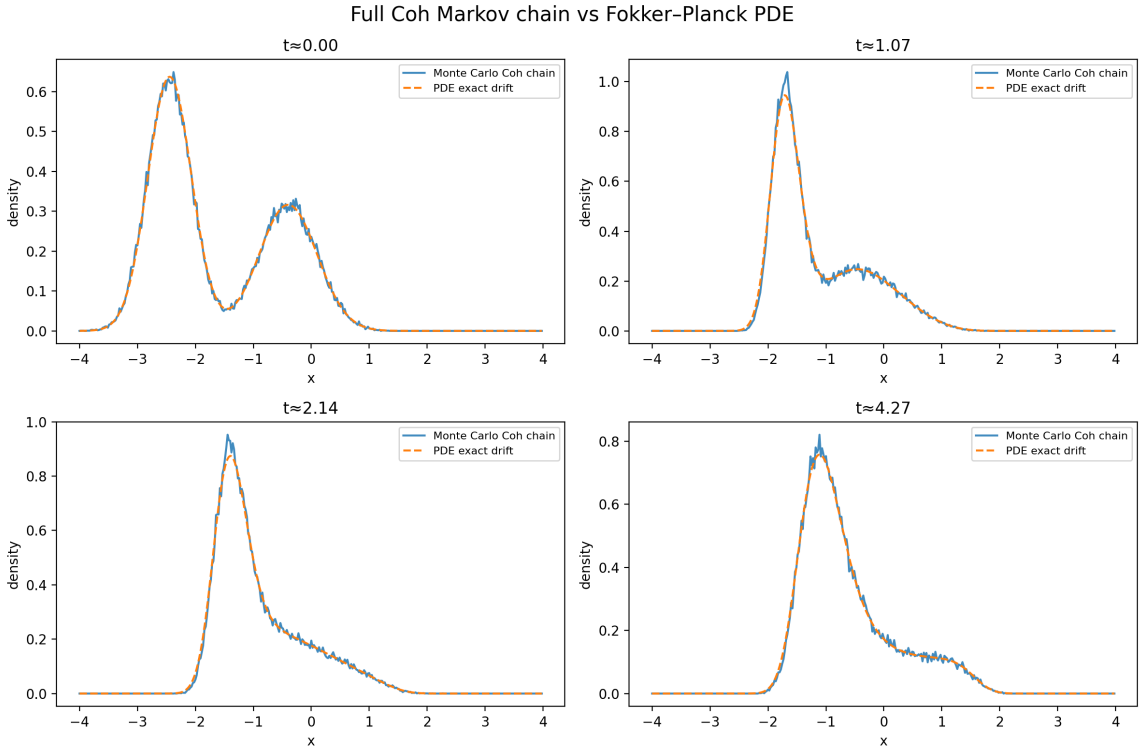

Figure 1: Discrete Coh Markov chain versus the Fokker–Planck density evolution in the full utility-plus-violation drift setting.

9 Discussion

The Coh probability law is not a generic softmax. It is a transition measure with two coupled layers:

$$\text{utility drive} \quad \times \quad \text{admissibility gate.}$$

In the hard limit, admissibility truncates the transition set. In the soft limit, the process becomes a governed stochastic flow. In the continuous limit, this produces the drift decomposition (16).

This construction is useful for systems where actions should not be selected solely by reward, utility, or gradient descent. A candidate transition must also satisfy a margin expressing cost, risk, uncertainty, or violation. Thus the law is naturally suited to governed computation, safe control, constrained optimization, and verification-aware agents.

10 Conclusion

We defined the Coh probability law, proved that it forms a valid probability measure, derived its Markov kernel, computed the generator, and established the Fokker–Planck limit under diffusive scaling. The resulting drift balances utility ascent against violation descent. Numerical validation confirms that the discrete Coh process and the derived PDE agree at the macroscopic density level. The construction provides a mathematically explicit bridge from admissibility-certified transitions to continuous governed stochastic dynamics.

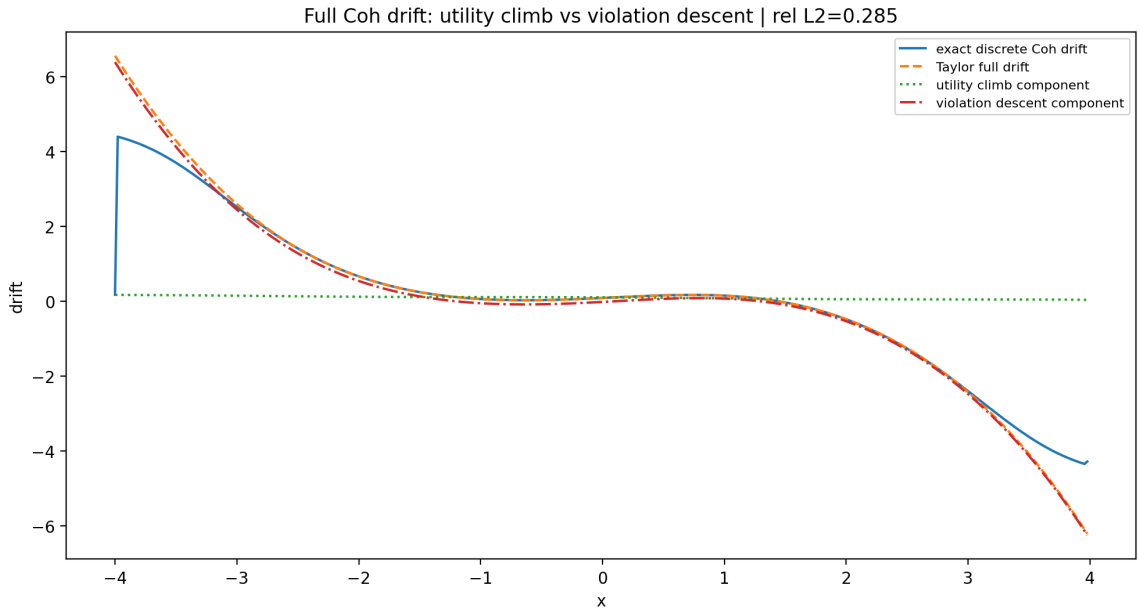

Figure 2: Exact discrete drift, Taylor drift, utility ascent component, and violation descent component.

References

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